

Abstract. Variational quantum algorithms are a leading route to simulating lattice field theories on near-term devices, but much of their cost is spent escaping poor parameter initializations. Q-MAML trains a classical network to predict circuit parameters across a family of Hamiltonians, yet has only been validated on spin chains and small molecules. We provide a from-scratch reimplementation that reproduces the original paper’s convergence gains (15–20× fewer iterations on H_2) and benchmark it on high-energy-physics (HEP) classification and on ten lattice and field-theory Hamiltonians, each verified before use and replicated over three seeds. Q-MAML attains the highest HIGGS classification accuracy at every qubit count tested. On the Hamiltonians the advantage is conditional: consistent wins on two of ten systems, seed-dependent wins on two, ties on four, a tie-to-slight-loss on one, and a loss on penalty-enforced $\mathbb{Z}_2$ gauge theory. That loss is an ansatz artifact: with a structurally gauge-invariant ansatz Q-MAML is the best scheme at all three seeds. At larger sizes, Q-MAML’s failures trace to pretraining converging to poor local minima of the meta-objective, visible in the training loss without test data, and restarting from the lowest-loss of several Learner initializations recovers a win. A quantum-Fisher-information probe shows Q-MAML initializations sit at higher effective dimension than random ones (0.93–0.96 versus 0.68–0.74, 4–12 qubits), consistent with a benefit from placement in parameter space rather than resistance to barren plateaus.

Keywords: variational quantum eigensolver, meta-learning, barren plateaus, lattice gauge theory, high-energy physics, quantum machine learning

Learned initialization for variational quantum circuits on lattice and field-theory Hamiltonians: a ten-system benchmark of Q-MAML

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1 Introduction

Simulating lattice field theories is a flagship application for quantum computers, and the variational quantum eigensolver (VQE) is the workhorse for near-term ground-state preparation. Its cost is the number of circuit evaluations, and a large share is spent escaping poor initial parameters: random initializations are prone to barren plateaus [3, 4], and the initialization heuristics proposed to avoid them, such as near-identity blocks [5], Gaussian [6] or reduced-domain [7] sampling, are independent of the Hamiltonian being solved.

A complementary idea is to *learn* the initialization. Meta-VQE [2] learns energy profiles of parameterized Hamiltonians; Q-MAML [1] trains a classical “Learner” network that maps a task descriptor (the Hamiltonian’s parameters) directly to circuit angles, by back-propagating the energy through the circuit with no inner adaptation loop; recent work generates initial parameters with graph networks [8] or diffusion models [9]. Q-MAML was validated on Heisenberg chains and small molecules only, and whether it helps on the Hamiltonians and datasets of high-energy physics (HEP) is open.

We make three contributions. **(i)** We implement Q-MAML from the published algorithm and validate it against the original paper’s own settings before using it on anything new, alongside a first-order variant with a truncated inner loop and a stop-gradient update as a further classification baseline. **(ii)** We benchmark it on HEP classification and on ten lattice and field-theory Hamiltonians, spanning Abelian and non-Abelian gauge theories, scalar and fermion–boson field theories, supersymmetric quantum mechanics, nuclei and neutrinos; every Hamiltonian is verified against exact limits before use and every comparison is replicated over three seeds. **(iii)** We show that the outcome is conditional on ansatz structure and system size, diagnose two of the mechanisms, and give a diagnostic that flags a failing initialization from the training loss alone. Code, results and notebooks are public (Data and code availability).

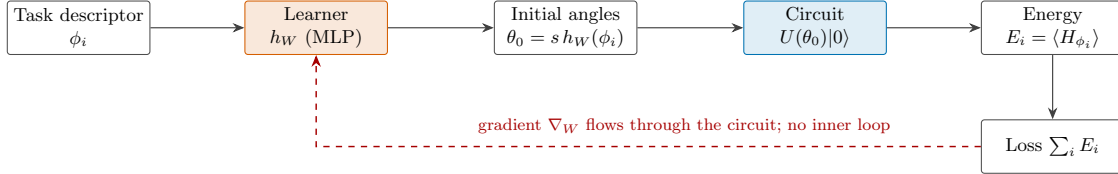
2 Methods

2.1 Q-MAML

A *task* is a Hamiltonian H_ϕ with descriptor $\phi \in \mathbb{R}^d$ (masses, couplings, a discrete label). A parameterized circuit $U(\theta)$ on n qubits prepares $|\psi(\theta)\rangle$ and the task cost is $\ell_\phi(\theta) = \langle \psi(\theta) | H_\phi | \psi(\theta) \rangle$. Q-MAML (Fig. 1, Algorithm 1) has two phases. *Pretraining* fits a Learner h_W , a two-layer MLP

(width 256, double precision), so that $\theta_0 = s h_W(\phi)$ ($s = 0.3$) has low energy on training tasks, by minimizing $\sum_i \ell_{\phi_i}(\theta_0^{(i)})$ over W with Adam. Gradients flow through the circuit (adjoint or back-propagation) and there is no inner loop. *Adaptation* freezes W and, on a held-out task, runs gradient descent on ℓ_{ϕ^*} starting at $\theta_0 = s h_W(\phi^*)$. Any initializer can replace the Learner in the second phase, so the comparison isolates initialization quality.

a Pretraining (Algorithm 1): learn to map a task to good angles



b Adaptation (Algorithm 2): a standard VQE loop from the predicted start

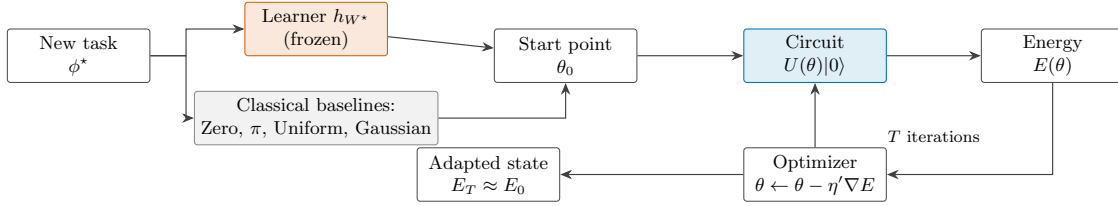


Figure 1: **The Q-MAML framework.** **a**, Pretraining (Algorithm 1): a Learner network maps each training task’s descriptor to initial circuit angles; the energy is back-propagated through the circuit (dashed red path) to update the Learner, with no inner loop. **b**, Adaptation (Algorithm 2) on a held-out task: the frozen Learner (or a classical baseline) supplies θ_0 , and a standard VQE loop refines it. All initializers share the same adaptation phase.

Baselines. All baselines feed the same adaptation phase: Zero ($\theta = 0$), π ($\theta = \pi$), Uniform ($\theta \sim \mathcal{U}[-\alpha\pi, \alpha\pi]$, $\alpha = 0.05$) and Gaussian ($\theta \sim \mathcal{N}(0, \gamma^2)$, $\gamma^2 = 1/(4S(L+2))$, $S = 2$, L layers) [6]. For classification we add a first-order MAML variant: several inner gradient steps on the circuit parameters on each support set, followed by a query loss evaluated at $w_0 + \text{sg}(w_T - w_0)$, where sg is the stop-gradient operator and w_T the parameters after T inner steps. It shares the Learner-free circuit and head of the other classification arms, so it separates the effect of the outer-loop objective from that of the architecture.

2.2 Benchmarks

Classification. HIGGS (tabular features, 180 000-event subsample [10]) with tasks defined as quantile bins of the reconstructed m_{bb} (six bins, 8 support and 8 query events per task, 18 training and 12 test tasks); a hybrid model (feature encoder, parameterized circuit, linear head), depth 2, ten epochs, 4/6/8 qubits. A second dataset, quark–gluon jet tagging, is run at six qubits.

Hamiltonians. Fig. 2 and Table 1 list the ten systems. Each has its own module with exact diagonalization, a task sampler and a verification routine (Hermiticity; exact analytic limits such

Algorithm 1 Q-MAML pretraining and adaptation

Require: task distribution $p(T)$, ansatz $U(\theta)$, Learner h_W , scale s , learning rates η, η'

Pretraining (Algorithm 1)

- 1: **for** each epoch **do**
- 2: **for** each training task $T_i \sim p(T)$ **do**
- 3: $\theta_0^{(i)} \leftarrow s h_W(\phi_i); \quad E_i \leftarrow \langle \psi(\theta_0^{(i)}) | H_i | \psi(\theta_0^{(i)}) \rangle$
- 4: $W \leftarrow W - \eta \nabla_W E_i$ $\triangleright$ through the circuit; no inner loop
- 5: **end for**
- 6: **end for**

Adaptation (Algorithm 2) on a held-out task T^*

- 7: $\theta \leftarrow s h_W(\phi^*)$
 - 8: **for** $t = 1, \dots, T$ **do** $\theta \leftarrow \theta - \eta' \nabla_\theta \ell_{\phi^*}(\theta)$
 - 9: **end for**
-

as the harmonic-oscillator spectrum, the $V=0$ limit or the $\mathfrak{su}(2)$ algebra; symmetry commutation, e.g. $[H, G_L] = 0$ and $G_L^2 = \mathbb{K}$ for the $\mathbb{Z}_2$ Gauss operators; and, for $\mathbb{D}_8$, a representation-theory check over all 64 group-element pairs that caught a real bug before any run). Two systems needed a stated change of question: the neutrino and Yukawa sources study real-time dynamics, so we ask for the ground state of the same Hamiltonian instead.

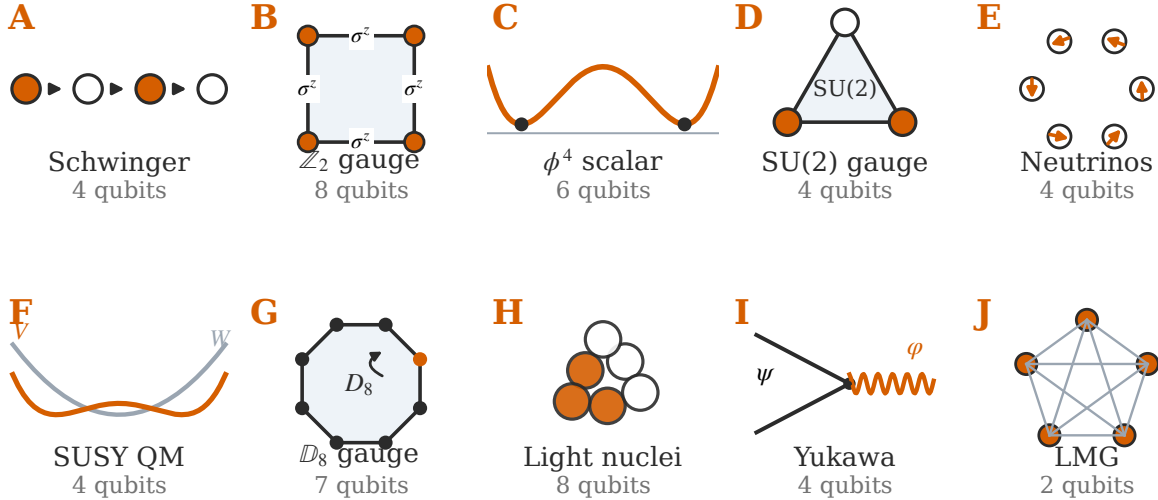


Figure 2: **The ten benchmark Hamiltonians.** Track letters as in Table 1; qubit counts are the base sizes.

2.3 Protocol and metrics

For each system, 16 training and 6 held-out tasks are drawn from the descriptor range with disjoint random seeds (held-out *interpolation*); the Learner is pretrained for 40 epochs and each scheme is adapted on each held-out task for 150–300 Adam iterations. We report the final relative energy error $\varepsilon = \exp(\log |E_T - E_0|_{\text{tasks}} / \langle |E_0| \rangle_{\text{tasks}})$ against the exact ground energy E_0 , or the log-gap $\log |E_T - E_0|$ where $E_0 \approx 0$ (supersymmetric quantum mechanics). Every comparison is repeated for three seeds, which change the training and test tasks, the Learner initialization and

Table 1: **The ten benchmark Hamiltonians.** HEA: hardware-efficient ansatz; HVA: Hamiltonian-variational ansatz; UCCSD: unitary coupled-cluster singles and doubles. Qubits in parentheses are the larger sizes run as extensions.

Track	System (source)	Qubits	Ansatz	Task parameters
A	Lattice Schwinger model, 1+1D QED [11]	4 (6, 8)	HEA	(m_0, g)
B	Single-plaquette $\mathbb{Z}_2$ LGT [12]	8	HEA / HVA	(J, m, μ)
C	ϕ^4 scalar field [13]	6	HEA	(m_0^2, λ_0)
D	SU(2) LGT, gauge field eliminated [14]	4 (6, 8)	HEA	$(\tilde{m}, x)$
E	Collective neutrino oscillations [15, 16]	4 (6)	HEA	(θ, μ)
F	Supersymmetric quantum mechanics [17]	4	HEA	(superpotential, m, g, μ)
G	Non-Abelian $\mathbb{D}_8$ LGT [18]	7	HEA	(M, λ_E, ϵ)
H	Light nuclei, lattice pionless EFT [19]	8	UCCSD	(nucleus, t, C, D)
I	Scalar Yukawa coupling [20]	4	HEA	(η, m)
J	Lipkin–Meshkov–Glick model [21]	2 (3)	HEA	(ϵ, V)

the baselines’ random draws. A seed is scored as a Q-MAML *win* or *loss* if its error is more than 10% below or above the best classical scheme at that seed, and a *tie* otherwise, including when both errors are below 10^{-6} (the floor set by the 10^{-8} regularizer in the log-gap). Classification uses Welch’s t -test and Cohen’s $d = (\bar{x}_1 - \bar{x}_2)/s_p$ with $s_p = \sqrt{(s_1^2 + s_2^2)/2}$, uncorrected for multiple comparisons unless stated.

3 Results

3.1 Validation against the original paper

Before extending Q-MAML we checked that our implementation reproduces the original paper’s own settings (Fig. 3). On the H_2 molecule, with the paper’s exact circuit (4 qubits, depth 7, 2000 iterations), Q-MAML reaches machine precision by iteration ~ 100 whereas classical initializations need 1500–2000, a 15–20 $\times$ reduction in circuit evaluations, and all non-degenerate schemes converge to the same final energy. On the Heisenberg XYZ chain (6 qubits; the paper’s circuit is not available to us, so the ansatz is reconstructed) the gain is 4–6 $\times$. In both, the advantage is convergence speed from a better starting point, not a different optimum.

3.2 HEP classification

On HIGGS, Q-MAML has the highest mean validation accuracy at every qubit count, over 8 seeds each (Fig. 4, Table 2). Table 2 gives the tests against Gaussian initialization and against the post-hoc best classical scheme; the margin against Gaussian is significant at every qubit count ($p < 0.01$, $d > 2.0$), and against the best classical scheme it is significant at 4 and 6 qubits ($p < 0.02$, $d > 1.5$) and directional at 8 ($p = 0.081$, $d = 0.96$). Q-MAML is also the only variant whose training meta-loss decreases appreciably. On quark–gluon tagging (six qubits, two seeds) the gap is larger, 0.776 ± 0.005 against 0.594–0.599 for the classical schemes, but two seeds is directional evidence only.

Classification cost functions are also far more plateau-prone than the VQE costs: the gradient variance decays by ~ 8 orders of magnitude from 4 to 10 qubits for every initialization, to a

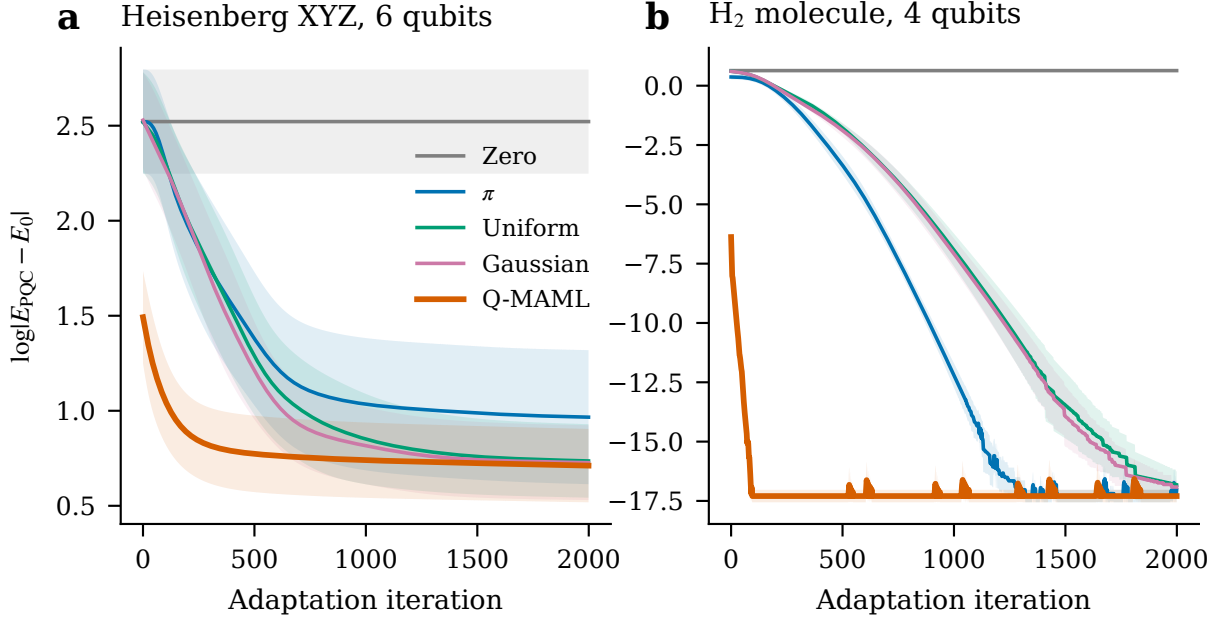


Figure 3: **Validation on the original paper’s settings.** Mean log-gap over held-out tasks (bands: half a standard deviation) versus adaptation iteration for **a**, the Heisenberg XYZ chain (reconstructed ansatz) and **b**, the H₂ molecule (exact ansatz). Q-MAML starts closer to the ground state and converges 4–6 $\times$ and 15–20 $\times$ faster, respectively.

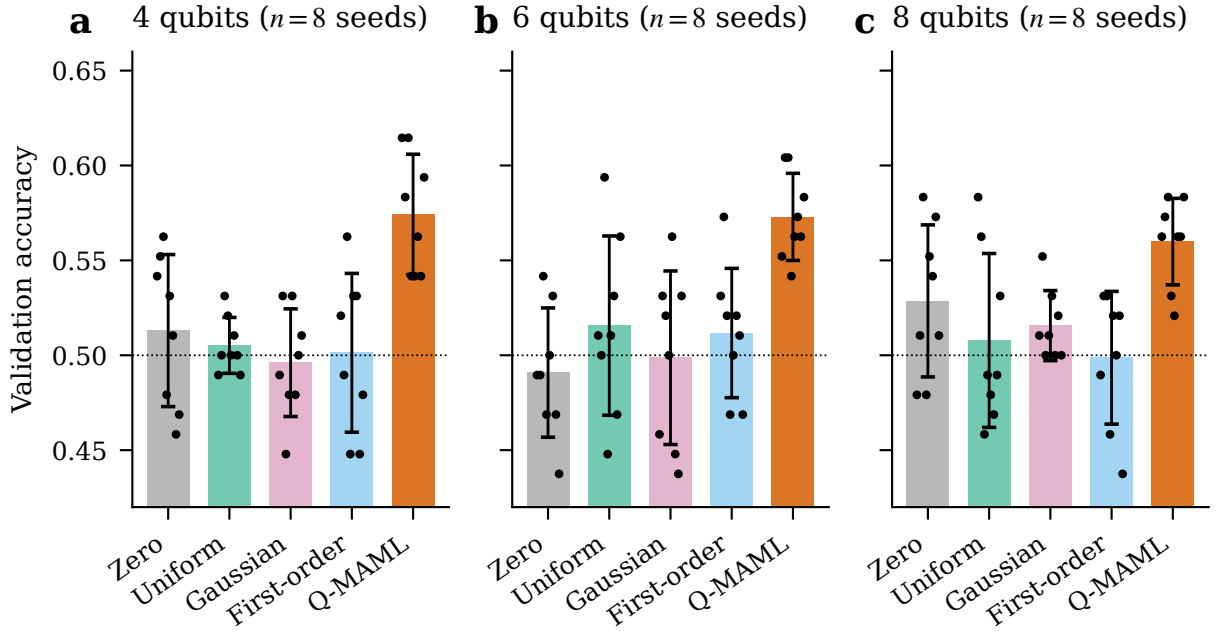


Figure 4: **HIGGS classification.** Final validation accuracy by initialization (bars: mean; error bars: standard deviation; dots: individual seeds). Q-MAML is highest at 4, 6 and 8 qubits.

common floor, consistent with cross-entropy behaving as a global cost [4], whereas the Heisenberg

Table 2: **HIGGS validation accuracy** (mean $\pm$ s.d. over n seeds), and Welch’s p (Cohen’s d) for Q-MAML against Gaussian and against the post-hoc best classical scheme (uncorrected).

	4 qubits	6 qubits	8 qubits
Seeds (n)	8	8	8
Zero	0.513 ± 0.040	0.491 ± 0.034	0.529 ± 0.040
Uniform	0.505 ± 0.015	0.516 ± 0.047	0.508 ± 0.046
Gaussian	0.496 ± 0.028	0.499 ± 0.046	0.516 ± 0.018
First-order variant	0.501 ± 0.042	0.512 ± 0.034	0.499 ± 0.035
Q-MAML	0.574 ± 0.032	0.573 ± 0.023	0.560 ± 0.023
Welch p (d), Q-MAML vs Gaussian	< 0.001 (2.6)	0.002 (2.1)	< 0.001 (2.1)
Welch p (d), vs best classical	0.005 (1.7; Zero)	0.011 (1.5; Uniform)	0.081 (1.0; Zero)

VQE cost shows no exponential decay over 4–10 qubits at depth 3. Q-MAML does not change the scaling exponent in either domain (Section 3.6).

3.3 Ten Hamiltonians

Table 3 and Fig. 5 summarize the ten systems at their base sizes. Two systems are consistent wins: the Schwinger model at $L=2$ (Q-MAML 0.18% against π ’s 0.22%; two wins and a tie), and $SU(2)$ gauge theory at $N=2$, the largest margin observed, where Q-MAML reaches $\sim 10^{-7}$ relative error at every seed against 10^{-4} for the best classical scheme, a $10^3\times$ margin. Neutrino oscillations ($N=4$) and supersymmetric quantum mechanics are wins at two of three seeds and a loss at the third, so the sign of the result is seed-dependent. Four systems are ties: the ϕ^4 scalar field (Q-MAML 6.05%, Zero 6.12%; in the single-seed convergence curves Q-MAML reaches this value about three times sooner), light nuclei (all five schemes agree within 0.03% at every seed, so the UCCSD ansatz makes the initial point irrelevant to the final energy, while Q-MAML alone avoids a shared oscillatory transient), scalar Yukawa (Q-MAML and Zero tie at the precision floor; the task range is near-vacuum, where Zero already starts close), and the Lipkin–Meshkov–Glick model (all schemes at or near the precision floor, and still at $J=3.5$, so Hilbert-space size alone does not make it discriminating). The $\mathbb{D}_8$ theory is a tie to slight loss against Uniform and Gaussian. Finally, $\mathbb{Z}_2$ gauge theory with a Gauss-law penalty is a loss at every seed; Section 3.4 shows this is a property of the ansatz.

3.4 Structural gauge invariance decides the $\mathbb{Z}_2$ result

With Gauss’s law enforced by an energy penalty $V \sum_l (G_l - q_l)^2$ ($V=10$) on a hardware-efficient ansatz, Q-MAML loses to the best classical scheme at every one of three seeds ($49.4 \pm 13.7\%$ against 19.4% for Gaussian, the best classical mean) with a nearly flat adaptation curve. A sweep over the penalty strength $V \in \{1, 3, 10, 30\}$ refuted the obvious explanation that a large penalty drowns the task signal: Q-MAML is behind the best classical scheme at every V tested, tying it only at the weakest ($V=1$). Two further experiments isolate the cause (Fig. 7). Replacing the ansatz with a gauge-invariant Hamiltonian-variational ansatz, whose layers Trotterize the physical Hamiltonian’s own terms so that gauge invariance is structural, makes Q-MAML the best scheme at all three seeds, $3\times$ – $10\times$ ahead of the best classical scheme ($0.0022 \pm 0.0011\%$

Table 3: **Ten Hamiltonians at base size** ($n = 3$ seeds): mean final relative error of Q-MAML and of the best classical scheme (best mean), and per-seed wins/ties/losses of Q-MAML against the best classical scheme at each seed (10% tolerance; both below 10^{-6} counts as a tie). Track F is in log-gap because its ground energies are ≈ 0 .

Track	System	Q-MAML error (%)	Best classical (%)	W/T/L
A	Schwinger, $L=2$	0.181	0.216 (π)	2/1/0
B	$\mathbb{Z}_2$ LGT, penalty ansatz	49.4	19.4 (Gaussian)	0/0/3
B	$\mathbb{Z}_2$ LGT, gauge-inv. ansatz	2.2×10^{-3}	0.011 (Zero)	3/0/0
C	ϕ^4 scalar field	6.05	6.12 (Zero)	0/3/0
D	SU(2) LGT, $N=2$	1.3×10^{-5}	0.013 (Gaussian)	3/0/0
E	Neutrinos, $N=4$	3.6×10^{-3}	2.1×10^{-3} (π)	2/0/1
F	SUSY QM	-4.48 (log-gap)	-4.75 (Uniform)	2/0/1
G	$\mathbb{D}_8$ LGT	10.8	10.7 (Uniform)	0/1/2
H	Light nuclei (UCCSD)	1.44	1.44 (Gaussian)	0/3/0
I	Scalar Yukawa	2.7×10^{-5}	2.7×10^{-5} (Zero)	0/3/0
J	LMG, $J=3/2$	2.7×10^{-6}	1.7×10^{-6} (Zero)	0/3/0

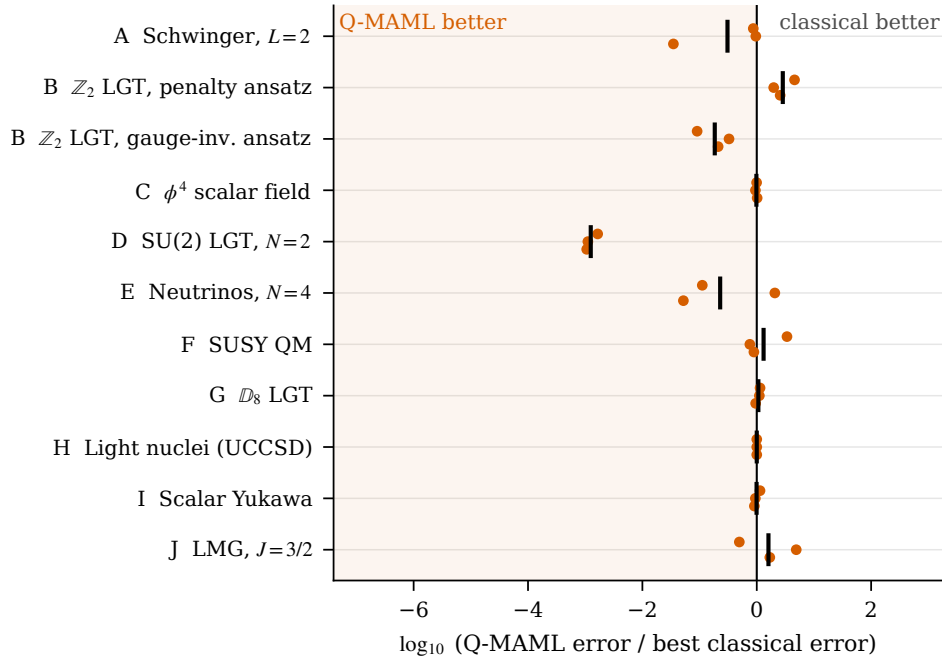


Figure 5: **Q-MAML against the best classical initialization, per seed.** Each dot is one seed (Q-MAML error divided by the best classical scheme’s error at that seed, log scale); black ticks are seed means. Left of zero, Q-MAML is better. Track B is shown with both ansatzes (Section 3.4).

against 0.011%). A second ansatz of similar cost that is *not* gauge-invariant leaves Q-MAML at or behind Uniform and Gaussian. Annealing the penalty during pretraining (Fig. 7b) roughly halves Q-MAML’s mean error on the original ansatz (49% $\rightarrow$ 22–32%) but beats every classical

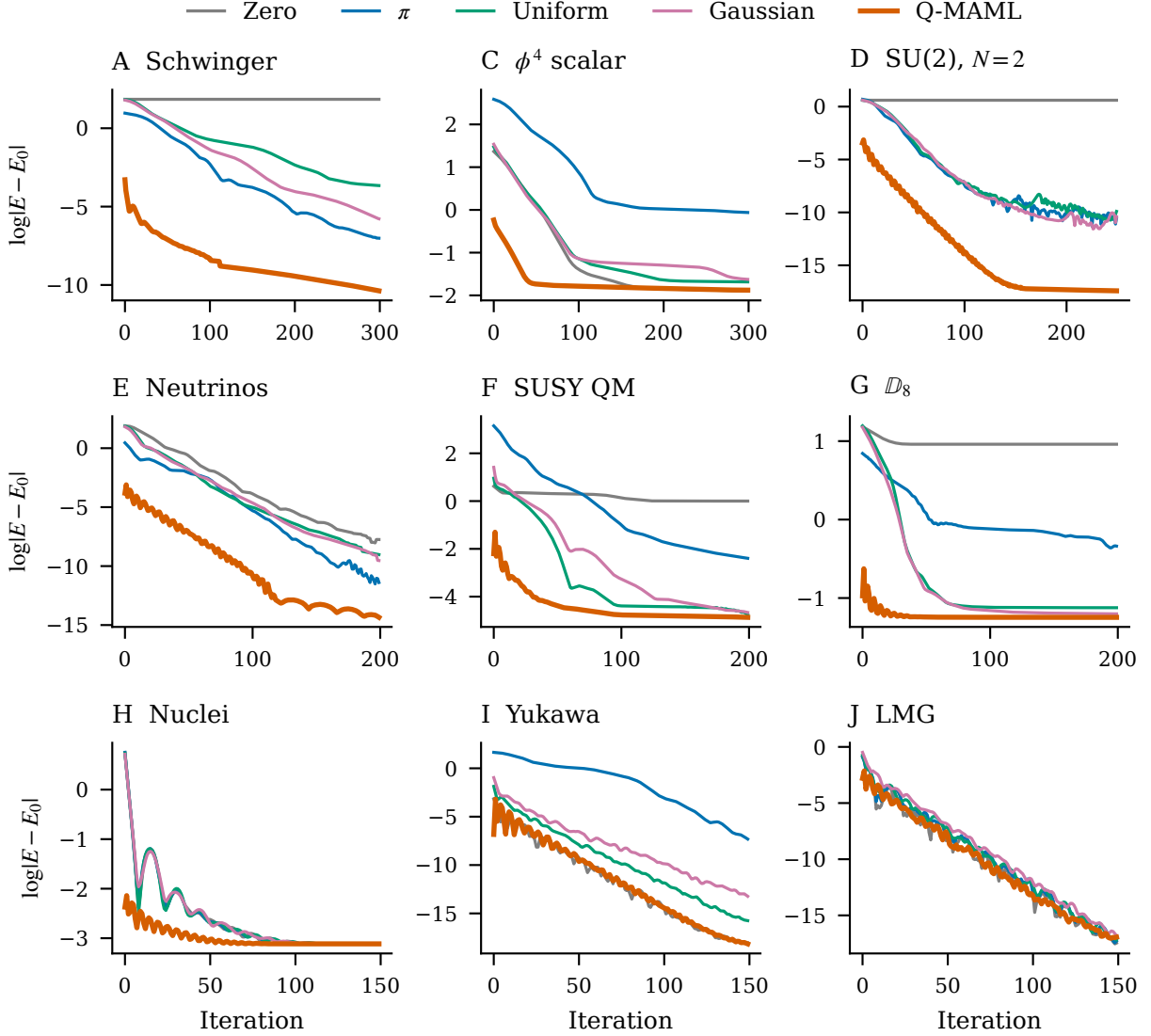


Figure 6: **Adaptation trajectories.** Mean log-gap to the ground energy over held-out tasks, by initialization, for the nine base-size systems not shown in Fig. 3. Track B is shown separately in Fig. 7.

scheme in only one of six seed-schedule cells. The final pretraining loss tracks the outcome: at $V=10$ the ground energies are $\approx +74.5$ (a penalty offset), the gauge-invariant Learner ends at the ground energy (74.6), and the annealed hardware-efficient Learners end 5 or 25–27 above it. These three comparisons use a reduced 4-task, 200-iteration budget; at the full 6-task, 300-iteration budget used everywhere else, the gauge-invariant ansatz replicates (1.4×10^{-5} against 4.3×10^{-5} for the best classical scheme, a win at all three seeds, 2.5–6 $\times$ ahead) and the control ansatz is a loss at two of three seeds and a tie at the third. The two ansätze are different published circuits rather than a minimal pair, so the best-supported reading is that structural, not penalty-enforced, gauge invariance is what lets a learned initialization work; it is not a controlled ablation.

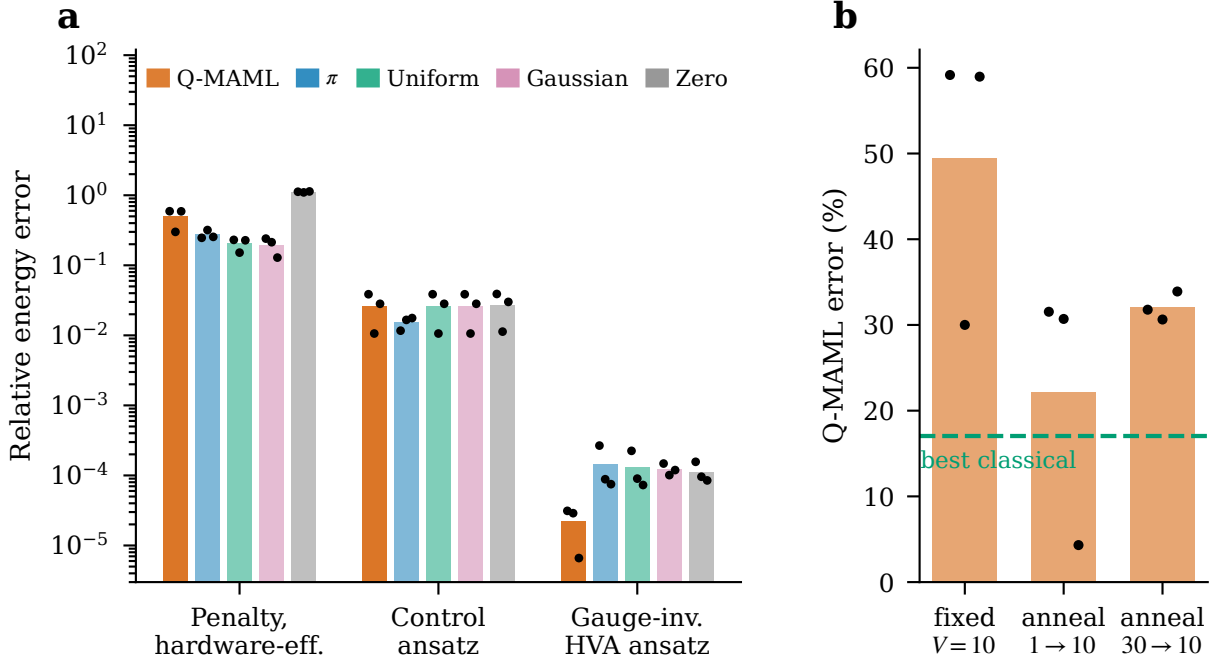


Figure 7: $\mathbb{Z}_2$ gauge theory: the ansatz decides. **a**, Relative error for each scheme (bars: mean; dots: seeds) on the penalty-enforced hardware-efficient ansatz, a control ansatz that is not gauge-invariant, and the gauge-invariant ansatz. **b**, Q-MAML error with the penalty fixed or annealed during pretraining, against the best classical scheme (dashed).

3.5 Behavior with system size

We extended three systems to larger sizes, verifying each Hamiltonian first (Fig. 8). For neutrino oscillations the win strengthens: at $N=6$ Q-MAML is best at all three seeds, $135\times$, $\approx 77\times$ and $144\times$ ahead of π , the best classical scheme. For $SU(2)$ gauge theory the $N=2$ advantage does not carry over. At $N=3$ a converging budget (depth 6) puts π first at every seed ($0.28 \pm 0.07\%$) with Q-MAML a stable second ($2.02 \pm 0.39\%$); at $N=4$ Q-MAML becomes seed-unstable, at 23.3%, 2.9% and 450% across three seeds while π holds at 3.5–6.0%. (An earlier depth-4 run at $N=3$ showed every scheme failing at 200–500%, a budget artifact.) For the Schwinger model the original $L=4$ (8-qubit) run failed, with Q-MAML at 31.8% against π ’s 8.6%.

Why Q-MAML fails at larger sizes. For $SU(2)$ $N=4$ we recorded the full pretraining history and per-task energies before and after adaptation for three seeds and two pretraining settings (Fig. 9a,b). The three seeds are three different regimes. At seed 1 both settings converge to the same good solution (training meta-loss -0.27 , initial energies within ~ 0.2 of the ground energies), and the adapted energies agree to 10^{-4} or better. At seed 2 both converge to the same bad one: the meta-loss stalls near 0.95 against ground energies of ≈ -0.3 , and adaptation cannot move the initialization at all (final equals initial energy to two digits), so the predicted point is effectively a stationary point of each task’s cost. At seed 0 the original setting diverges (the meta-loss rises) and a lower learning rate stalls on an intermediate plateau. A lower learning rate and more epochs, the fix that repaired the Schwinger $L=4$ pretraining, therefore does not repair these seeds. Two consequences follow: “pretraining converged” is not “pretraining

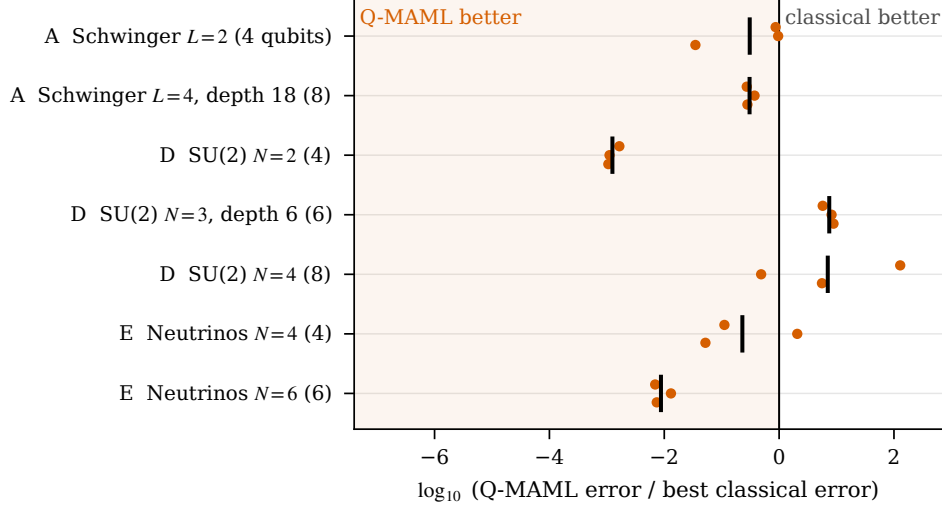


Figure 8: **Larger systems.** As in Fig. 5, for each system size (qubits in parentheses). The neutrino win strengthens with size; the SU(2) advantage at $N=2$ is replaced by a tie or loss and, at $N=4$, seed instability. The Schwinger $L=4$ point uses a lighter protocol (3 tasks, 150 iterations).

worked”, and the failure is visible in the training meta-loss without any test data. We tested the second consequence directly: six Learner initializations were pretrained per data seed at the original $N=4$ setting and scored by held-out error against final training loss (Fig. 9c). A final loss near the mean ground energy (≈ -0.3) was always followed by a low error (five of five restarts, 2.0–3.4%, at or below π); a stalled loss ($\gtrsim 1$) was a warning but not a reliable one, since one data seed’s stalled restarts spanned 2.0–15.1%. Selecting the restart with the lowest training loss beat π at both data seeds tested (2.3% and 2.0% against 3.5% and 4.2%), turning the diagnostic into a usable selection rule. The Schwinger model at $L=4$ behaves differently. A lower pretraining learning rate repairs pretraining (mean energy inside the range of the true ground energies) but at depth 12 the adapted result stays mixed across seeds (21.2% against π ’s 17.9% at seed 0; 20.4% against 25.0% at seed 1), whereas depth 18 gives Q-MAML the lead at every seed run (Fig. 8), even at the original learning rate where pretraining still stalls (seed 0: 4.5% against 8.7%). Depth, not pretraining convergence, is the operative variable there; whether a stalled-pretraining Learner still contributes beyond a small-angle start was not isolated.

3.6 Placement in parameter space, not plateau resistance

A McClean-style scan of the gradient variance at a reference parameter shows Q-MAML’s variance comparable to or larger than the classical schemes’ at every qubit count from 4 to 14 and no change in the scaling exponent, so Q-MAML is not resisting barren plateaus. A single-coordinate gradient cannot distinguish a circuit that is degenerate everywhere from one degenerate along a few directions, so we also computed the quantum Fisher information matrix (block-diagonal metric tensor) at each sampled initialization and summarized it by the participation ratio of its spectrum, normalized by the number of parameters, an effective dimension (Fig. 9d). Q-MAML initializations have consistently higher effective dimension (0.925 ± 0.022 at 4 qubits rising to 0.959 ± 0.011 at 12) than Uniform (0.684–0.695) or Gaussian (0.721–0.741), and the gap does not

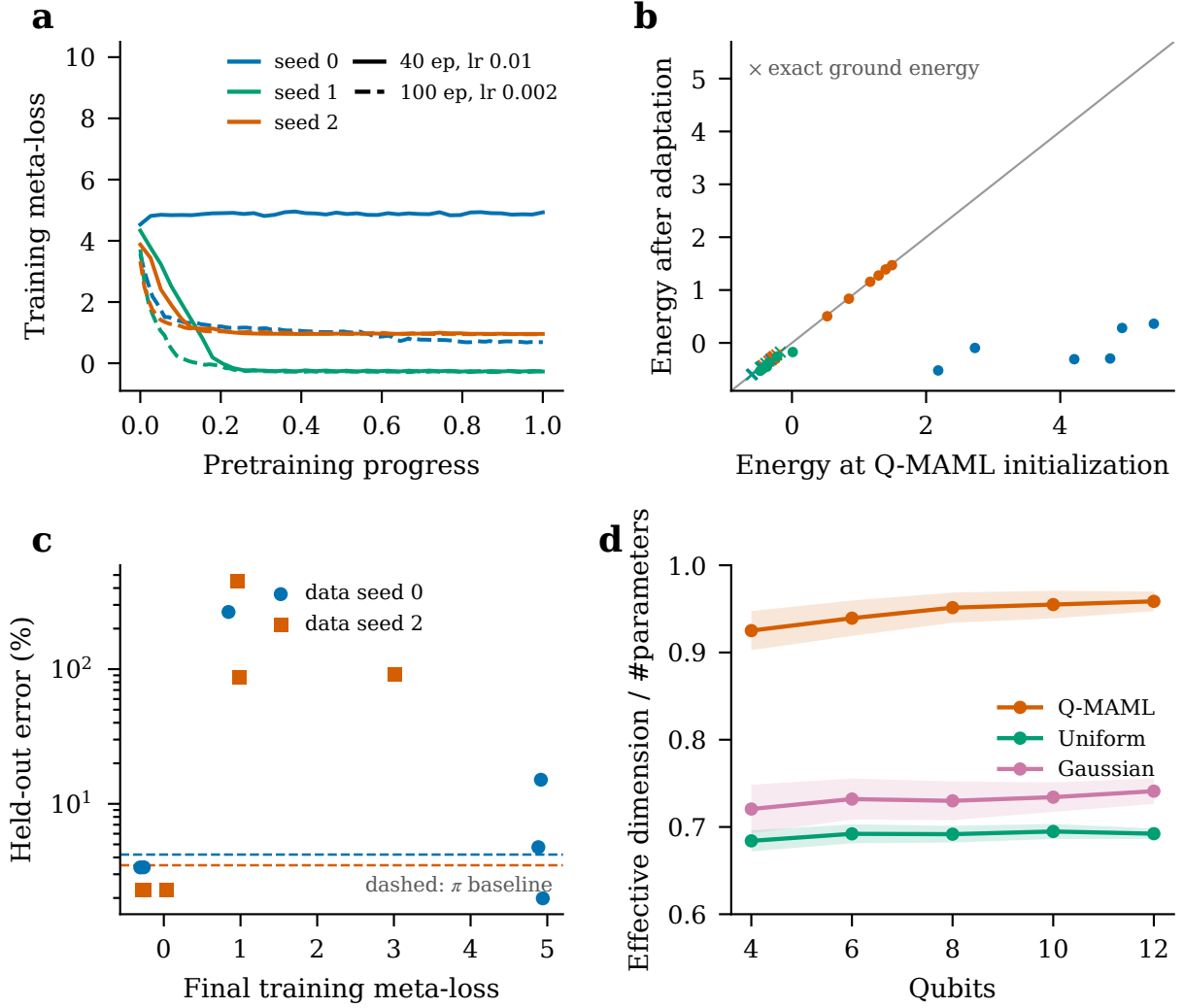


Figure 9: **Mechanisms.** **a**, $SU(2)$ $N=4$ training meta-loss for three seeds and two pretraining settings: seed 1 converges to a good solution, seed 2 to a poor one, seed 0 diverges at the original setting. **b**, Energy at Q-MAML’s initialization against energy after adaptation, per test task (crosses: exact ground energies): at seed 2 points lie on the diagonal (adaptation cannot move them), far above the ground energies. **c**, Held-out error against final training loss for six Learner restarts at each of two data seeds; dashed lines are the π baseline. **d**, Effective dimension of the quantum Fisher information matrix at initialization (Heisenberg XYZ ansatz, 20 samples per point; shading: s.d.).

shrink with qubit number. This gives quantitative form to the intuition that Q-MAML’s benefit lies in where it places the circuit in parameter space; it measures local effective dimension at the initialization, on one ansatz, not whether that translates into faster adaptation.

4 Discussion

The results do not support a blanket claim that learned initialization helps HEP simulations; they support a conditional one. It helps where a task-dependent good region exists and pre-training can find it: small systems with structurally symmetric ansätze (Schwinger $L=2$, $SU(2)$ $N=2$, gauge-invariant $\mathbb{Z}_2$), and neutrino oscillations up to six qubits. It is unnecessary where the initial point does not matter (UCCSD nuclei), where a trivial initialization is already near the answer (near-vacuum Yukawa) or where the benchmark is too easy to discriminate (LMG). And it is unreliable where the meta-objective has poor local minima that pretraining finds ($SU(2)$ $N=4$). Two practical recommendations follow. Compare against π and Uniform, since no classical scheme is a safe default (π is best at some systems and worst at others) and the π baseline was best or tied for best at several sizes. And monitor the training meta-loss against the (cheap, classically computed) mean ground energy of the training tasks: when several Learner initializations are pretrained, the one with the lowest final loss is a usable selection rule, since it recovered a win over π at $SU(2)$ $N=4$ without test data.

Limitations. The largest system is 12 qubits (the Fisher probe) and typically 4–8; all results are noiseless simulation of interpolation over a task range, and we do not test extrapolation across a phase transition beyond one scalar-field experiment where Q-MAML did not generalize. Three seeds resolve large effects but not small ones, and several extensions (neutrino $N=6$ excepted) and mechanism experiments use a reduced budget or a single seed, as stated. Track-level results use a fixed hardware-efficient ansatz (except nuclei and the $\mathbb{Z}_2$ comparison), so they measure relative initialization quality on a given ansatz, not the best achievable VQE. We use the unreduced $SU(2)$ and $\mathbb{D}_8$ Hamiltonians, not the qubit-reduced constructions of their source papers. Track E and I are reformulations of real-time problems as ground-state problems. The $SU(2)$ $N=4$ instability is diagnosed, not fixed.

5 Conclusion

Q-MAML reproduces the original paper’s 15–20 $\times$ gains and improves HEP classification, but on lattice and field-theory Hamiltonians its benefit is conditional on ansatz structure and system size, and two apparent results reverse under scrutiny: a loss on $\mathbb{Z}_2$ gauge theory that is an artifact of the penalty-enforced ansatz, and an $SU(2)$ advantage at four qubits that does not survive to eight. The most transferable findings are diagnostic: a failing initialization is visible in the training meta-loss and can be worked around by restarting and selecting on that loss, and a Fisher-information probe shows Q-MAML’s gain is placement in parameter space, not resistance to plateaus. Natural next steps are larger systems and hardware noise, and the qubit-reduced formulations of the $SU(2)$ and $\mathbb{D}_8$ theories.

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Declarations

- **Funding:** [to be confirmed by the author]
- **Competing interests:** [to be confirmed by the author]

- **Data and code availability:** All code, notebooks and saved results are at [repository URL to be completed by the author]. The HIGGS dataset is public [10].
- **Author contribution:** A.K. designed and ran all experiments and wrote the manuscript.

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